

Johannes Hosle

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EMPLOYMENT

Carnegie Mellon University in Qatar Postdoctoral Fellow in Mathematics Mentors: Layan El Hajj and Henrik Shahgholian	2026–present
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EDUCATION

Massachusetts Institute of Technology Ph.D. in Mathematics Advisor: David Jerison	2020–2026
University of California, Los Angeles B.Sc. in Mathematics with Departmental Honors, <i>summa cum laude</i> M.A. in Mathematics	2016–2020

PUBLICATIONS AND PREPRINTS

J. Hosle, **Monotonicity formulas in positively curved settings with applications to two-phase free boundary problems**, submitted.

J. Hosle, **The isomorphic section-projection problem for convex bodies**, *Bull. Lond. Math. Soc.*, to appear.

J. Hosle and P. Ivanisvili, **Geometric Block Exponents and a Uniform Mixed-Alphabet Sumset Inequality**, submitted.

J. Hosle, **Stolarsky-Type Inequalities in a Max-Convolution Problem**, submitted.

J. Hosle, A. V. Kolesnikov, and G. V. Livshyts, **On the L_p -Brunn–Minkowski and dimensional Brunn–Minkowski conjectures for log-concave measures**, *J. Geom. Anal.* (2021).

J. Hosle, **On extensions of the Loomis–Whitney inequality and Ball’s inequality for concave, homogeneous measures**, *Adv. Appl. Math.* (2020).

J. Hosle, **On the Comparison of Measures of Convex Bodies via Projections and Sections**, *Int. Math. Res. Not. IMRN* (2021).

HONORS AND AWARDS

Sigma Xi	2026
Dean of Science Fellowship, MIT	2020–2023
Phi Beta Kappa	2020
Sherwood Prize, UCLA	2020
Dean’s Honors List	2016–2020
Putnam Competition, Top 250	2016
National Merit Scholarship Semifinalist	2015

TEACHING AND GRADING

18.095 Mathematics Lecture Series, MIT Teaching Assistant (J. Dunkel)	Jan. 2026
18.065 Matrix Methods in Data Analysis, Signal Processing, and Machine Learning, MIT Teaching Assistant (Z. Chen)	Spring 2025
18.100B Real Analysis, MIT Course Assistant (L. Guth)	Fall 2024
18.100P Real Analysis, MIT Recitation Leader (K. Naff)	Spring 2024
18.01 Calculus, MIT Recitation Leader (L. Guth)	Fall 2023

MENTORSHIP EXPERIENCE

Mathroots @ MIT

July 2026

Academic mentor in two-week mathematical talent accelerator for high school students hosted by MIT PRIMES.

MIT Directed Reading Program

Jan. 2025

Mentored undergraduates Ryan Dong and Caleb Pascale in convex geometry, including the Busemann–Petty problem and Bourgain’s bound for the isotropic constant.

PRESENTATIONS

Analysis Seminar, Columbia University

Sept. 2026

Analysis Seminar, Brown University

Sept. 2026

Synergies between Geometry, Probability, and Computation in High Dimensions

Aug. 2026

ICERM, Brown University, Providence, Rhode Island

Workshop in PDE and Applied Math in the Northeast Region

Mar. 2026

University of Connecticut, Storrs, Connecticut

Pure Math Graduate Student Seminar, MIT

Feb. 2025

BIRS Workshop: Geometric Tomography

Feb. 2020

BIRS, Banff, Canada

Workshop in Convexity and Geometric Aspects of Harmonic Analysis

Dec. 2019

Georgia Institute of Technology, Atlanta, Georgia

RESEARCH PROGRAMS AND SUMMER SCHOOLS

Interactions between Convex Geometry and Spectral Analysis

July 2025

Université de Montréal, Montréal, Canada

Particle Interactive Systems: Analysis and Computational Methods

June 2024

SLMath, Berkeley, California

Georgia Tech REU in Mathematics

Summer 2018

Supervisors: Michael Lacey and Galyna Livshyts

PROGRAMMING

Python: Proficient; background in statistical learning theory; experience with NumPy and scikit-learn.

Julia: Proficient; used for gradient flows in free boundary problems, supersolution construction, and numerical estimates for weighted Wirtinger inequalities.

Wolfram Language: Intermediate; familiar with machine learning methods.

LaTeX: Proficient.

LANGUAGES AND CITIZENSHIP

Languages: English (native), Russian (elementary).

Citizenships: United States and Germany.